



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Affiliated to DBATU, RTMNU & MSBTE, Mumbai.
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An Autonomous Institute, with NAAC "A" Grade
Department of Computer Science & Engineering
"A place to learn; A chance to grow"



Session: 2025-26

MSE-1 EXAMINATION

Course: B. Tech. in Computer Science & Engineering

Year/Semester: 5th Semester (3rd Year)

Subject: DBMS

Subject Code: CS5T001

Max Marks: 20

Date: 04/08/2025

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Learn and understand fundamentals of database management system.
2. Exhibit the query development knowledge.
3. Learn modelling and normalization of databases.
4. Learn query processing and optimization techniques.
5. Analyse the File Organization, Indexing and Hashing and exhibit the knowledge of transaction and concurrency control.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the term attribute.	L1/CO1	1
Q.2	Classify different types of data models.	L1/CO1	1
Q.3	List the various relational algebra operators.	L1/CO2	1
Q.4	State the advantages of DBMS.	L1/CO1	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Discuss about applications of Database Systems?	L2/CO1	2
Q.6	Explain the term DDL and DML commands in SQL.	L2/CO2	2
Q.7	Differentiate the relational algebra and calculus.	L4/CO2	2
Q.8	Write the syntax for alter a table in SQL?	L1/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Describe the tier architecture of DBMS.	L2/CO1	4
Q.10	Give Symbol used in E-R Diagram and Draw the E-R diagram of University Management System.	L5/CO1	4

Consider the below two tables for reference to solve the SQL queries:

Q.11	i) Write an SQL query to fetch the EmpId and FullName of all the employees working under the Manager with id – '986'.	L3/CO2	4
	ii) Write an SQL query to fetch those employees who live in Toronto and work under the manager with ManagerId – 321.		
	iii) Write an SQL query to fetch common records between two tables.		
	iv) Write an SQL query to fetch all the EmpIds which are present in either of the tables – 'EmployeeDetails' and 'EmployeeSalary'.		

Table – EmployeeDetails

EmpId	FullName	ManagerId	DateOfJoining	City
121	John Snow	321	01/31/2014	Toronto
321	Walter White	986	01/30/2015	California
421	Kuldeep Rana	876	27/11/2016	New Delhi

Table – EmployeeSalary

EmpId	Project	Salary	Variable
121	P1	8000	500
321	P2	10000	1000
421	P1	12000	0